

Underwater superoleophobic composite coating characteristic of durable antifouling and anticorrosion properties in marine environment

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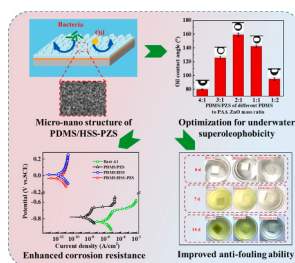
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HIGHLIGHTS

- Underwater oil-repellent, waterproof and anticorrosion coating was prepared by “Glue + Powder” strategy.
- The composite coating showed excellent antifouling and anticorrosion properties in 3.5% NaCl solution.
- The antifouling and anticorrosion is attributed to the superoleophobic surface and superhydrophobic interlayer.

GRAPHICAL ABSTRACT



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ABSTRACT

Biofouling of marine vessels often results in low fuel efficiency and severe hull corrosion, an ecological friendly coating with long durability and high Benefit-cost is greatly desired to prevent biofouling and bioadhesion. Herein, a facile “Glue + Powder” strategy was employed to synthesize a composite coating with satisfactory oil-repellent and anti-corrosion properties, in which the hexadecyl-trimethoxysilane modified SiO₂ nanoparticles (NPs) act as superhydrophobic undercoating and polyacrylic acid modified ZnO NPs implanted in polydimethylsiloxane as superoleophobic finishing coat. Immersion tests evidenced that no obvious cracks appeared on the surface of the bifunctional composite coating, without apparent adhesion loss even after 30 d of immersion in 3.5% NaCl solution. Electrochemical tests demonstrated that the Al alloy electrode lacquered by the composite coating presents a high impedance (above 10⁹ Ω cm²) even after 25 d of immersion in 3.5% NaCl solution, dramatically reducing the corrosion rate of Al alloy by 7 orders of magnitude. In addition, tribocorrosion test indicated that the coated Al alloy specimen can survive from 120 cycles of friction-wear under 0.5 MPa loading, exhibiting excellent corrosion and wear resistance. Antifouling test showed that the bifunctional composite coating could reduce the underwater adhesion of oily molecules with an oil contact angle of 160 ± 2°, presenting excellent superoleophobicity and immune of the adhesion of *Pseudomonas* sp. and following formation of biofilm. Hopefully, it could be one of competitive marine antifouling coatings.

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1. Introduction

Marine biofouling is the undesirable attachment and accumulation of organics and microorganisms, plants, and their deposits on artificial material surfaces immersed in seawater [1]. For the global shipping business, marine biofouling is a nightmare that has plagued it for centuries [2–5]. The biofouling is prone to accumulate on the vessel hull, leading to sailing velocity loss and fuel efficiency reduction, even serious corrosion as Dabalà, M commented [6]. Antifouling coating is one of the most direct and effective methods to reduce bioadhesion. Traditionally, some toxic biocides such as tributyltin, zinc pyrithione, copper pyrithione, was widely employed in antifouling coatings to prevent the attachment of microorganisms. However, long-term accumulation of toxic substances may impose irretrievably harm on marine ecology. Therefore, eco-friendly antifouling technologies have attracted more and more attention, such as sol-gel coating, silicone coating, photocatalytic coating [10]. It's believed that the biofouling begins with the surface contamination and conditioning film formed by the adsorption of oily molecules, followed by the attachment and accumulation of microorganisms [7]. Hence, it is imaginable biofouling could be refrained through reducing the affinity of oily molecules on coating surface.

After long-term natural competition and selection [8], some creatures have evolved special abilities to adapt to environmental changes, and these peculiar morphologies and functions urge scientists to develop biomimetic materials [9–15]. For instance, the superwetttable surfaces mimicking *Nepenthes mirabilis* [16–18] have attracted lots of attention in marine antifouling researches [19], which can refrain the initial attachment of oil contamination on hull surfaces. In addition, it is found that the self-cleaning sharkskin is also attributed to the Superhydrophilic and Underwater Superoleophobic (SUS) sharkskin, which can decrease the contact area between oily substances and the sharkskin by capturing a water layer.

Recently, SUS surface has been energetically applied in marine biofouling. For example, Chen et al. reported a self-repairing underwater superoleophobic and antibiofouling coating through the self-assembly of hydrophilic polymeric chain modified hierarchical microgel spheres, and the obtained coating exhibited excellent underwater superoleophobic and antibiofouling properties [20]. Zarghami et al. fabricated a SUS coating through polymer surface engineering, in which the polymeric coating was prepared by zwitterionic phase separation polymeric membrane [21].

However, the stability of most traditional SUS coating is usually unsatisfactory in seawater, which is mainly attributed to the seawater absorption of coatings or mechanical damage on their surface [22]. To overcome the weakness of SUS coatings, superhydrophobic surfaces (SHS) have received much attention due to their peculiar blocking barrier on water and corrosive medium [23–28]. Selim et al. [29–32] reported a multifunctional superhydrophobic coating fabricated by ternary nanocomposites of polydimethylsiloxane (PDMS)/graphene oxide nanosheets decorated with ZnO nanorods [33], and the obtained superhydrophobic coating not only exhibit outstanding biofouling resistance, but also endures 500 h salt spray test. Nevertheless, it's worth noting that the mechanical stability and bondability of most artificial

SHS are less than satisfactory [34]. Even worse, the inherent micro-nano rough morphology on SHS surface may afford more colony sites and voids as bacterial reservoirs. Accordingly, how to exploit strengths and avoid weaknesses of SHS and SUS is prominently worthy of attention. Unfortunately, to our knowledge, limited works have been done about the synergistic application of SHS and SUS.

In this work, a “Glue + Powder” strategy [35] was employed to fabricate bi-layered coating by combining SHS and SUS. Firstly, SiO₂ nanoparticles (NPs) modified by eco-friendly hexadecyltrimethoxysilane (HDTMS) were physically blended with PDMS binder to construct SHS as undercoating, aiming at blocking corrosive species ingress and improving adhesion strength. Then, polyacrylic acid (PAA) was grafted on ZnO NPs (PAA-ZnO), and then blended with PDMS to act as antifouling finishing coat. the composite coating exhibits excellent anti-biofouling performance thanks to the synergetic effect of the oil-repellent SUS and the antibacterial ZnO NPs [36,37]. Moreover, the composite coating also presented promising corrosion resistance due to that the intimal SHS can inhibit the ingress of seawater and the underlying corrosion of the Al matrix.

2. Materials and methods

2.1. Materials

Different specimens including glass, aluminum, steel, poly (tetrafluoroethylene) (PTFE), wood and polyester fabric plates are used as the substrates. Polydimethylsiloxane (PDMS) and its curing agent, hexadecyltrimethoxysilane (HDTMS), polyacrylic acid (PAA; aqueous solution, 50%), SiO₂ nanoparticles (NPs, ~100 nm in diameter) were purchased from Aladdin Industrial Corporation (China), and ZnO particles, n-hexane (C₆H₁₄), sodium chloride (NaCl) and absolute alcohol (EtOH) from Sinopharm Chemical Reagent Co., Ltd (China).

2.2. Preparation of superhydrophobic HD-SiO₂ NPs

Generally, SiO₂ NPs contain a mass of hydroxyl and exhibit hydrophilicity, to obtain superhydrophobicity SiO₂, 0.2 g of SiO₂ NPs with an average size of 100 nm was dispersed in a hybrid suspension composed of 50 mL of EtOH, 50 mL of deionized water and 2 mL of hexadecyltrimethoxysilane (HDTMS) [38]. Then, the hybrid suspension was kept at 30 °C and stirred at 400 rpm for 12 h. Afterward, the suspension was centrifugally rinsed with ethanol 5 times, and then dried at 60 °C for 24 h. The obtained HD-SiO₂ was stored in a desiccator for later use.

2.3. Fabrication of superhydrophobic PDMS/HD-SiO₂ surface

Superhydrophobic PDMS/HD-SiO₂ surfaces were fabricated by spraying HD-SiO₂ suspension onto substrates using a spray gun with φ0.4 mm nozzle [38]. First, 0.5 g of HD-SiO₂ NPs were added into 30 mL of n-hexane, and then vigorously stirred at 500 rpm for 5 min and ultrasonically dispersed for 2 min. Subsequently, 1.0 g of PDMS and 0.1 g of curing agent were added into the above suspension and ultrasonically dispersed. The prepared PDMS/HD-SiO₂ suspension was sprayed onto substrates and cured at 100 °C for 2 h to obtain superhydrophobic PDMS/HD-SiO₂ coatings.

2.4. Fabrication of superoleophobic PDMS/PAA-ZnO surfaces

Superoleophobic PAA-ZnO surfaces were also fabricated by spraying method. First, 0.4 g of ZnO particles were dispersed in a mixture containing 25 mL of ethanol and 5 mL of deionized water, and then 1.0 g of PAA was added into the mixture. The formed suspension was stirred at room temperature for 2 h, followed by addition of a certain ratio of PDMS. Afterward, the suspension was sprayed onto the PDMS/HD-SiO₂ layer, and the as-prepared composite coating (named PDMS/HSS-PZS) was finally cured at 100 °C for 2 h. For comparison, the

Table 1
Abbreviation and properties of different coatings.

Samples	Description	Properties
HD-SiO ₂ NPs	HDTMS wrapped SiO ₂ NPs	Superhydrophobicity
PDMS/PZS coating	Monolayer of PDMS/PAA-ZnO	Superoleophobicity
PDMS/HSS coating	Monolayer of PDMS/HD-SiO ₂	Superhydrophobicity
PDMS/HSS-PZS coating	Composite coating with outer PDMS/PAA-ZnO and inner PDMS/HD-SiO ₂ layer	Outer superoleophobicity and inner superhydrophobicity

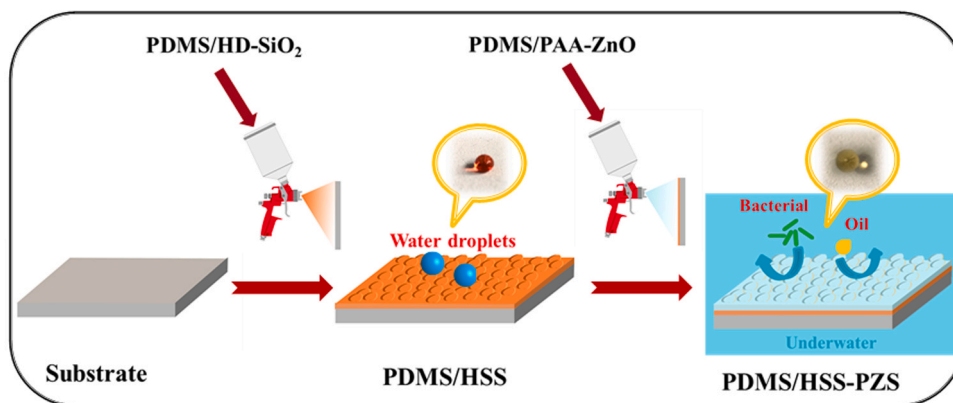


Fig. 1. Schematic diagram of the preparation process of underwater superoleophobic composite coating.

superoleophobic sample in the absence of PDMS/HD-SiO₂ layer and the superhydrophobic sample without the PDMS/PAA-ZnO layer were also prepared, abbreviated as PDMS/PZS and PDMS/HSS, respectively. Table 1 lists the abbreviation of all samples for clarity. Unless otherwise specified, AA6063 aluminum alloy [39] plate is used as the substrate in this work.

2.5. Characterization and electrochemical tests

A field-emission scanning electron microscope (FSEM, Gemini 300, Germany) was used to examine the morphologies of surface and cross-section. The micromorphology and chemical composition of HD-SiO₂ and PAA-ZnO particles were examined by transmission electron microscope (TEM, Tecnai G2 20, US) operating energy dispersive X-ray (EDX, X-Max, Oxford Instrument, UK) analysis. Atomic force microscopy (AFM, SPM-9700, Japan) was applied to measure the surface roughness of coatings with a scanning scale of 5 $\mu\text{m} \times 5 \mu\text{m}$. The thermal stability of PDMS/HSS and PDMS/PZS coating was evaluated using a thermal gravimetric analyzer (TGA, STA449F3, Germany) from 30 $^{\circ}\text{C}$ to 800 $^{\circ}\text{C}$ at a heating rate of 10 $^{\circ}\text{C}/\text{min}$ under the protection of N₂ gas. The adhesion of coatings was measured by a pull-off adhesion tester (PosiTest AT-A, U.S.). To evaluate the surface wettability, water and oil contact angles (WCAs and OCAs) were measured through an optical contact angle instrument (JC2000D, China). The chemical composition of different particles was determined by Fourier transforms infrared spectroscopy (FTIR, VERTEX 70, Germany).

Reciprocating friction and tape-peeling tests were carried out to evaluate the mechanical stability of the coating. Firstly, the fabricated

coatings were placed face-down on a piece of 1500# sandpaper and then were horizontally dragged along the sandpaper for a few cycles under 2 kPa pressure. Tribocorrosion behavior of the fabricated coatings was tested using a tribocorrosion tester (MS-ECT3000) at a friction rate of 10 cycles/min in 3.5 wt% NaCl solution under 0.5 MPa pressure, with a Si₃N₄ $\phi 6$ mm ball as the grinding head. All tests were performed three times with 3 parallel samples for reproducibility.

For electrochemical test, a three-electrode system was used to evaluate the corrosion behavior of uncoated and coated Al samples in 3.5 wt% NaCl solution, Pt mesh and saturated calomel electrode (SCE) as counter and reference electrodes, respectively. An AA6063 Al alloy disc was used as the working electrode, which was polished in sequence by 600 #, 1000 #, 1500 # emery paper, followed by ultrasonically rinsed with ethanol and distilled water, and finally dried in nitrogen flow. Tafel polarization curve and electrochemical impedance spectroscopy (EIS) were conducted via an electrochemical workstation (CS350, CorrTest, China). All working electrodes were pre-immersed in the NaCl solution for 30 min before each measurement until a stable open circuit potential (OCP) was observed. EIS was carried out in a frequency range of $10^5 \sim 10^{-2}$ Hz with a sinusoidal amplitude of 5 mV. Tafel polarization was conducted from -100 to 250 mV versus OCP at a scan rate of 0.5 mV/s. The obtained EIS and Tafel results were respectively analyzed with CS Studio software.

2.6. Antibacterial and antifouling tests

To assess the antibacterial performance, marine bacteria *Pseudomonas* sp. was selected as an experimental strain, and the antibacterial

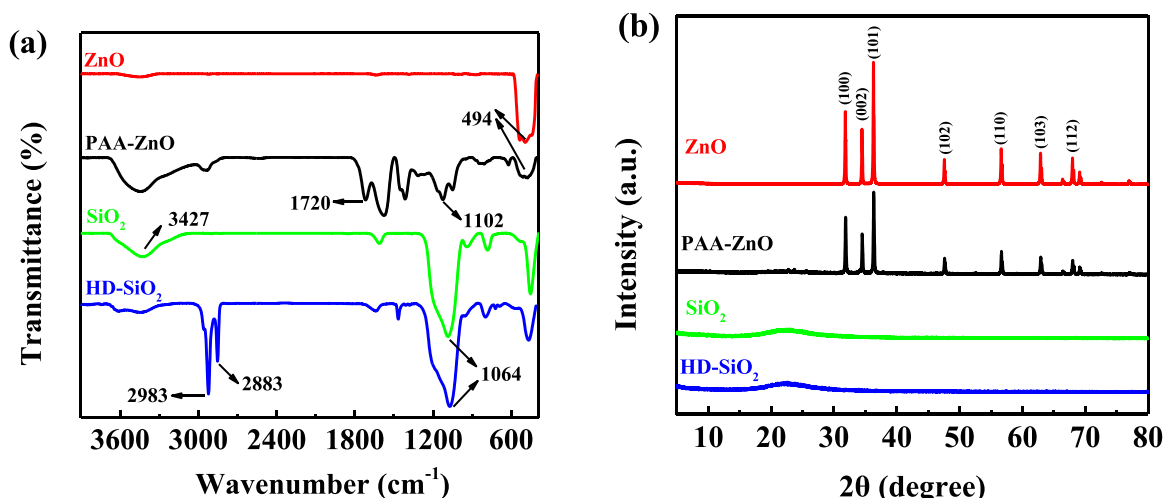


Fig. 2. (a) FTIR spectra and (b) XRD patterns of ZnO, PAA-ZnO, SiO₂ and HD-SiO₂ NPs.

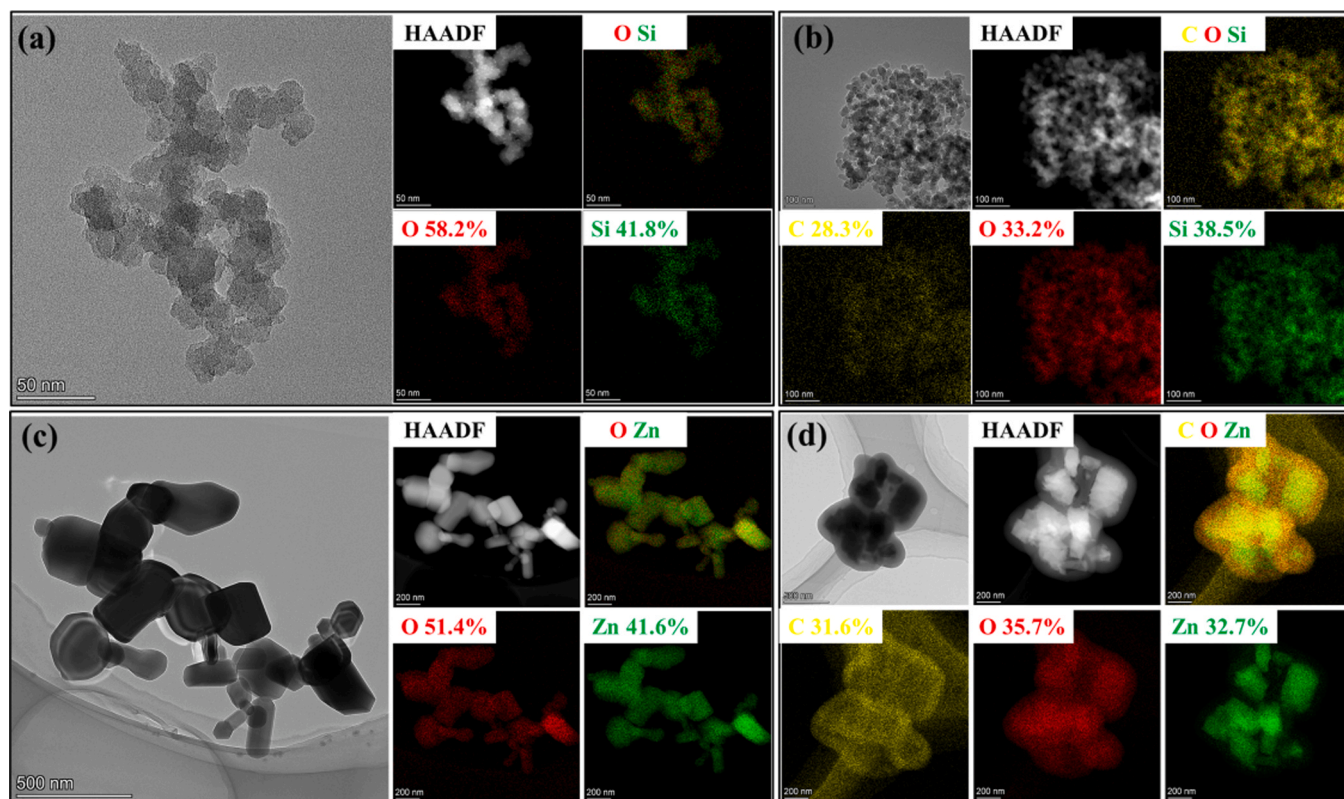


Fig. 3. TEM images and EDX elements mapping of (a) SiO₂, (b) HD-SiO₂, (c) ZnO and (d) PAA-ZnO NPs.

tests were carried out by plate colony-counting method according to Ref. [40]. Typically, different specimens were sterilized at high-temperature steam and then immersed into the bacteria suspension ($\sim 10^5$ cell/mL) at 28 °C for 7 d. Subsequently, the sessile bacteria were inoculated in a certain volume of culture medium and then diluted appropriately with phosphate buffer saline (PBS), 100 μ L of them was finally put into a solid culture plate. After 2 d of cultivation at 28 °C, the colonies on the specimens were counted, and their antibacterial effects were compared with the control specimen.

Chlorella vulgaris, taken from East Lake (Wuhan, China), was employed to evaluate the antifouling performance of the coatings. Firstly, the *Chlorella vulgaris* was cultured under laboratory conditions in open air at room temperature subjected to indirect natural light in daytime and darkness at night. The culture medium was composed of 0.0116 g/L CaCl₂·2H₂O, 0.24 g/L K₂HPO₄, 0.0153 g/L KH₂PO₄, 0.2 g/L MgSO₄, and 0.5 g/L carbamide referring to Ref. [41]. Afterward, *Chlorella vulgaris* was inoculated with 10% (v/v) seed culture. Coating specimens were put into the *Chlorella vulgaris* containing culture medium to observe the *Chlorella vulgaris* growth on different coating surfaces. To ensure the reproducibility of electrochemical, antibacterial and antifouling tests, three parallel samples were employed to obtain an average value in each test (Fig. 1).

3. Results and discussions

3.1. Chemical compositions analysis

FTIR was carried out to confirm the chemical compositions of different NPs including SiO₂, HD-SiO₂, ZnO and PAA-ZnO (Fig. 2a). For ZnO, the broad absorption band at 494 cm⁻¹ is ascribed to the characteristic absorption peak of Zn-O bands [42–44]. Compared with ZnO, two new absorption peaks appear at 1720 cm⁻¹ and 1102 cm⁻¹ in the FTIR spectrum of PAA-ZnO, which are attributed to the -COOH and -COO-, respectively [22], indicating that PAA was successfully grafted

on ZnO. For SiO₂, the strong peak at 1064 cm⁻¹ is ascribed to the characteristic absorption band of Si-O-Si [45], and the broad absorption band at 3427 cm⁻¹ is attributed to the stretching vibrations of -OH groups [38], which endows the pristine silica with hydrophilic performance. After being modified by HDTMS, two new adsorption bands at 2983 cm⁻¹ and 2883 cm⁻¹ are detected in the FTIR spectrum of HD-SiO₂, which are associated with the C-H stretching vibrations of the asymmetric methyl and symmetric methylene groups from HDTMS [26, 46]. Additionally, the disappearance of the adsorption peak at 3427 cm⁻¹ means that HDTMS has successfully modified the silica through the condensation reaction of Si-OH groups and HDTMS, which would benefit the superhydrophobicity. Moreover, the XRD patterns of ZnO, PAA-ZnO, SiO₂ and HD-SiO₂ NPs are shown in Fig. 2b. It can be found that the XRD patterns of SiO₂ and HD-SiO₂ exhibit a broad shoulder and have no characteristic sharp peaks, indicating the amorphous feature of SiO₂ NPs. For ZnO and PAA-ZnO, the diffraction peaks at 31.8°, 34.4°, 36.3°, 47.5°, 56.6°, 62.9° and 68.0° are related to the (100), (002), (101), (102), (110), (103) and (112) reflections of ZnO (PDF#89-0511).

Fig. 3 illustrates the TEM images and EDS elemental mapping of different NPs. It is observed that the morphologies of SiO₂ NPs before and after modification have no obvious change with the size of ~ 30 nm (Fig. 3a and b). Even so, the corresponding EDX mapping demonstrates that, in addition to existing Si and O elements, the HD-SiO₂ NPs are richer in C element than the pristine SiO₂ NPs, suggesting that HDTMS has been successfully grafted to the SiO₂ surface. Besides, Fig. 3c and d show that the ZnO NPs present rodlike feature with their surface covered by a thin film. The corresponding EDS manifests that C and O elements are distributed uniformly on the surface of ZnO, implying that the film is mainly composed of PAA.

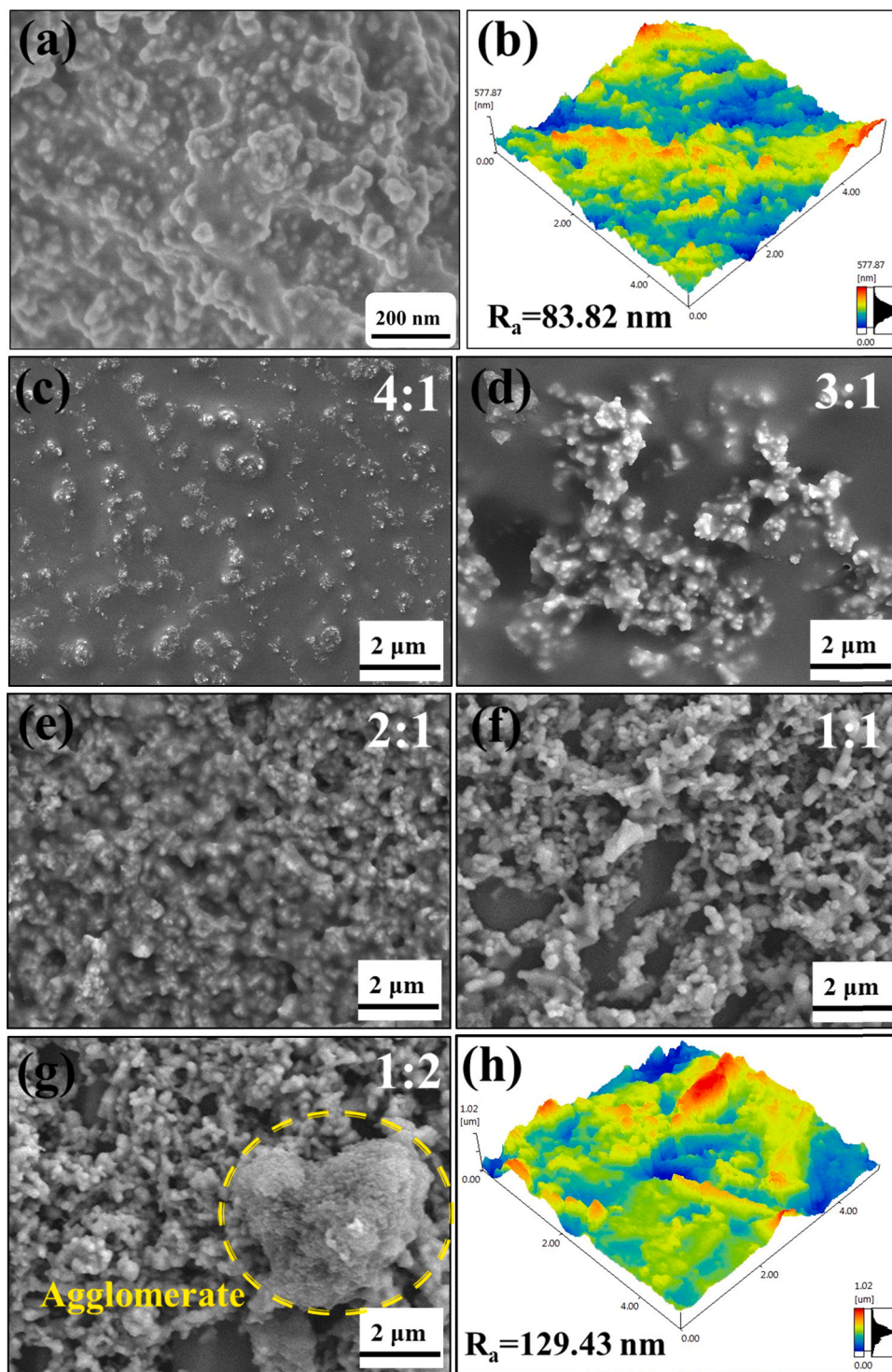


Fig. 4. SEM images of (a) PDMS/HSS and (c-g) PDMS/PZS at different PDMS to PAA-ZnO mass ratio and 3D AFM images of (b) PDMS/HSS and (h) PDMS/PZS.

3.2. Effects of the mass ratio of PDMS to PAA-ZnO on PDMS/PZS morphologies

It is well-known that the surface wettability of material is closely related to its microstructure [47–50]. Herein, the morphologies of PDMS/HSS and PDMS/PZS obtained by using different mass ratios of PDMS to PAA-ZnO were observed through SEM. As shown in Fig. 4a, the morphology of PDMS/HSS is featured by a lot of protuberances formed by HD-SiO₂. The PDMS grafted on HD-SiO₂ NPs could produce a rough surface through hierarchic micro/nanostructures. According to the AFM

image, PDMS/HSS demonstrates a bumpy structure with an average roughness of 83.82 nm (Fig. 4b). The surface morphologies of PDMS/PZS with different ratios of PDMS to PAA-ZnO are shown in Fig. 4c-g. When the mass ratio of PDMS to PAA-ZnO is 4:1, the PAA-ZnO NPs are almost completely soaked in PDMS (Fig. 4c), which is profitless for underwater superoleophobicity due to its relatively smooth or uniform surface. As expected, the surface roughness seems to increase with decreasing mass ratio of PDMS to PAA-ZnO (Fig. 4d-e), in which the PAA-ZnO NPs scatters more uniformly, generating a compact nanostructure. It is anticipated that such a hierarchical micro-/nanostructure

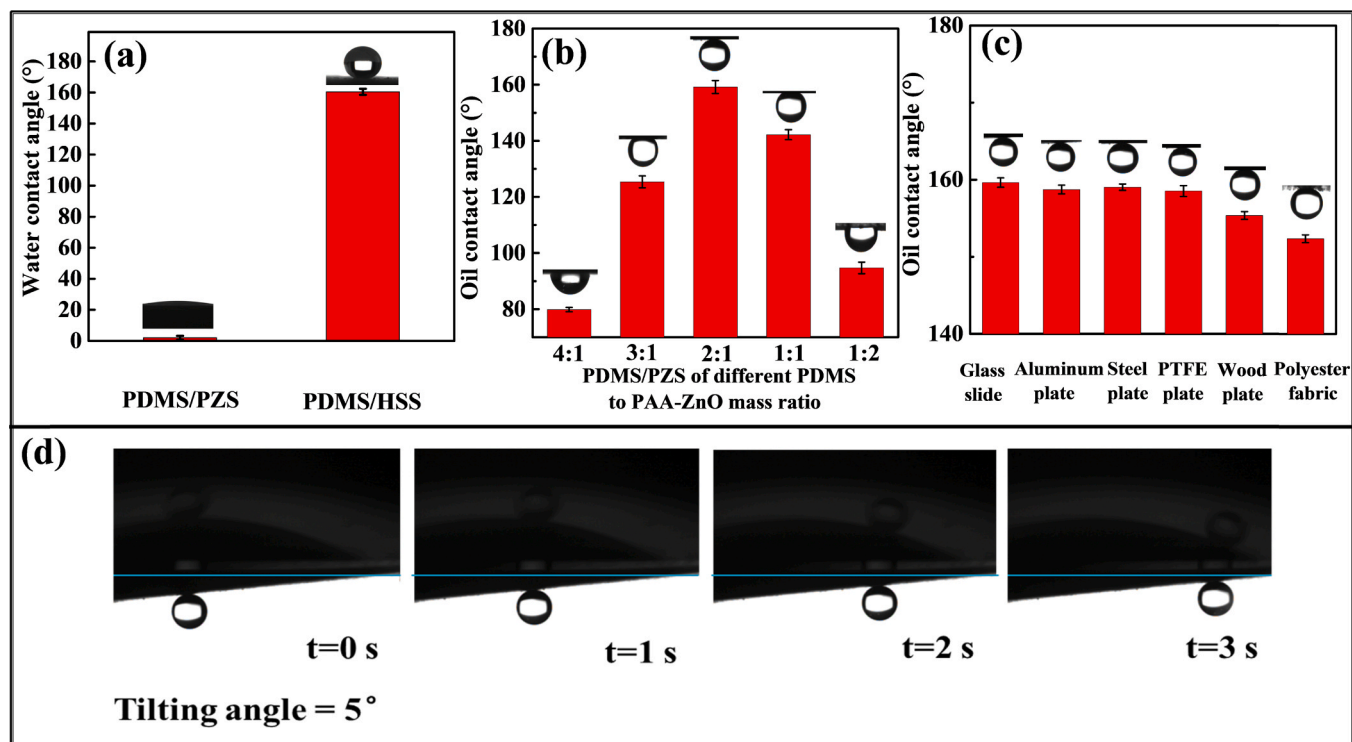


Fig. 5. (a) WCAs of PDMS/HSS and PDMS/PZS in air; (b) underwater OCAs of PDMS/PZS with different mass ratios of PDMS to PAA-ZnO; (c) underwater OCAs of PDMS/PZS on different substrates; (d) the rolling behavior of oil droplet on PDMS/PZS with a tilting angle of 5°.

would benefit their underwater superoleophobicity. In addition, the PDMS could strengthen the adhesion force between the PAA-ZnO layer and the underlying layer, which is beneficial to the mechanical stability of the composite coating. However, it is worth reminding that the PAA-ZnO NPs are distributed loosely and nonuniformly on the surface when the mass ratios of PDMS to PAA-ZnO decrease to 1:1 and 1:2 (Fig. 4f-g). Particularly, some PAA-ZnO NPs aggregates appear on the surface (Fig. 4g), implying that the PAA-ZnO NPs present poor adhesion strength to the underlying substrate when the mass ratio is too low, which is very detrimental to the stability of the PDMS/PZS surface and the underwater superoleophobicity. Therefore, the above results prove that a compact and stable PDMS/PZS surface could be achieved by the optimal mass ratio of PDMS and PAA-ZnO NPs (2:1). Moreover, the PDMS/PZS features a rugged micro-/nanostructure with an average roughness of 129.43 nm (Fig. 4h), facilitating their underwater superoleophobicity.

3.3. Effects of the PDMS to PAA-ZnO mass ratio on PDMS/PZS surface wettability

The effects of mass ratio of PDMS to PAA-ZnO on surface wettability of PDMS/PZS coating were investigated through underwater OCA measurements by using edible oil, as shown in Fig. 5. It is seen that the PDMS/PZS coating is superhydrophilic in air from Fig. 5a, however, the PDMS/HSS coating presents superhydrophobicity with a WCA of $160.4 \pm 2^\circ$, thanks to the synergistic effect of low surface energy hexadecyl groups and the micro-/nanostructured surface. Fig. 5b shows the underwater OCAs of PDMS/PZS with different mass ratios of PDMS to PAA-ZnO. The OCA of each specimen was measured after the specimen has been soaked in water for 30 min. It is seen that, as the ratio of PDMS to PAA-ZnO decreases from 4:1–1:2, the OCA increases first and then decreases, and the maximum OCA as high as $160.1 \pm 2^\circ$ is achieved when the mass ratio of PDMS to PAA-ZnO is 2:1. Neither excessive PDMS nor insufficient PDMS is favorable to underwater superoleophobicity. With the help of the compact micro-/nanostructures (Fig. 4a), a water

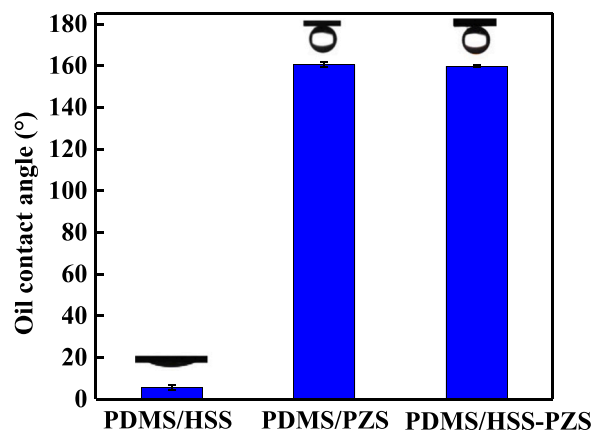


Fig. 6. Comparison of OCAs on different coating surfaces.

layer can be captured on the coating surface, which could significantly reduce the contact area between oil and the coating surface, thus effectively inhibiting the attachment of oil pollution underwater. Hence, according to the change of OCA, a proper mass ratio of PDMS to PAA-ZnO is the key to obtain a superoleophobic surface. unless otherwise specified, the mass ratio of PDMS to PAA-ZnO is optimized to 2:1 in the following tests.

In addition to AA6063, we also measured the OCAs of PDMS/PZS on other substrates, including glass, steel, PTFE, wood, and polyester fabric, as shown in Fig. 5c, all these surfaces are superoleophobic with underwater OCA $> 151^\circ$, indicating that the PDMS/PZS coatings are applicable to different substrates. Besides, when the substrate is tilted by 5°, the oil droplet can readily slide off the surface due to the buoyancy of water (Fig. 5d), indicating that oil droplets have extremely low affinity to the coating surface.

Furthermore, as shown in Fig. 6, we also measured the OCAs of the

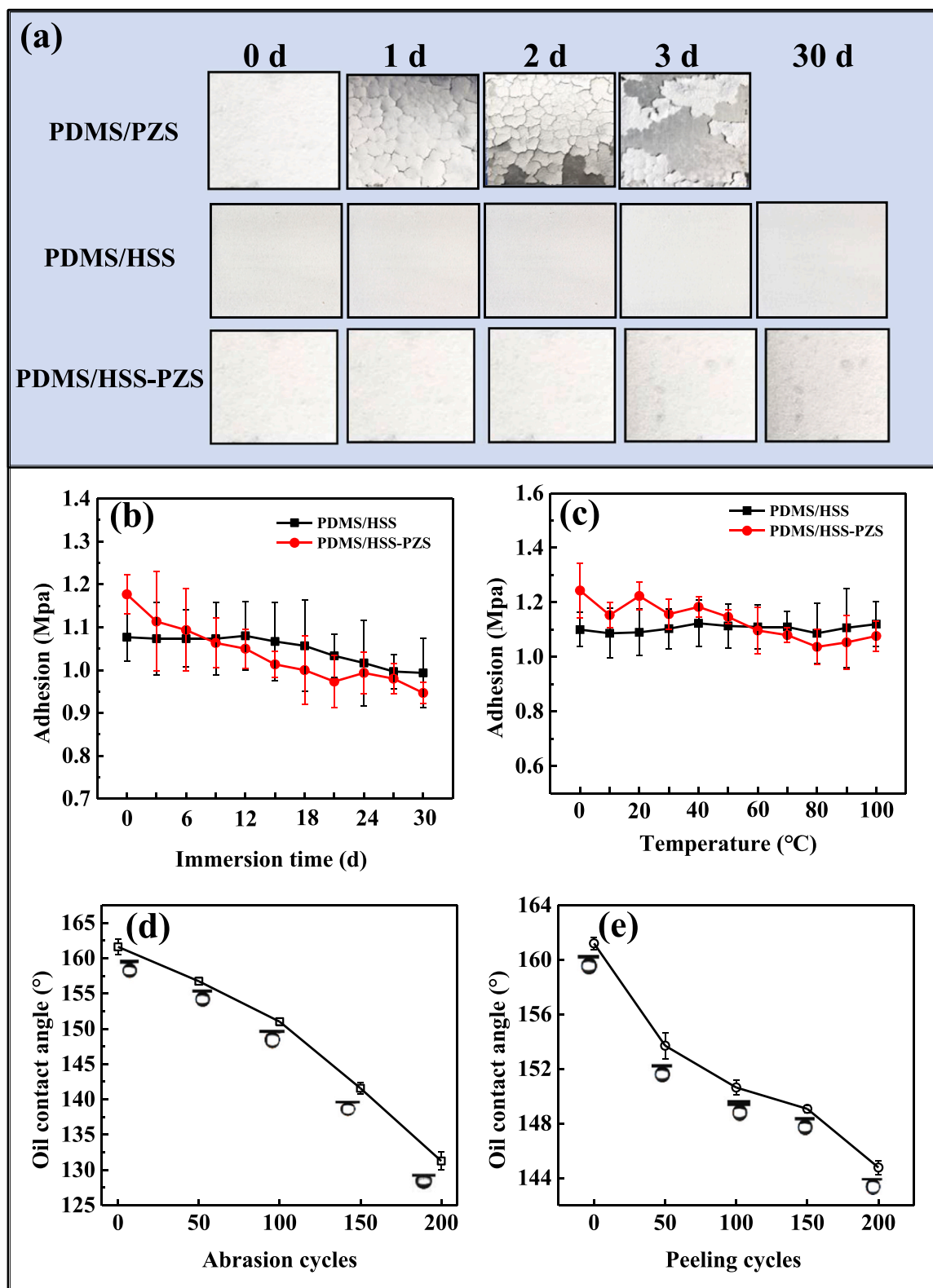


Fig. 7. (a) Morphologies of PDMS/PZS, PDMS/HSS and PDMS/HSS-PZS coatings versus immersion time in 3.5% NaCl solution; Adhesion strength of PDMS/HSS and PDMS/HSS-PZS coatings versus (b) immersion time and (c) temperatures; OCAs of PDMS/HSS-PZS coating versus (d) abrasion cycles and (e) peeling cycles.

PDMS/HSS and PDMS/HSS-PZS surfaces. It can be found that the PDMS/HSS surface is oleophilic and the oil droplets can spread out completely on it, apparently unable to prevent the adhesion of oil stains. Nevertheless, the PDMS/HSS-PZS surface becomes superoleophobic with an OCA of $160.2 \pm 2^\circ$ underwater, so does the PDMS/PZS surface.

The above results demonstrate that the PDMS/HSS surface has a relatively high affinity to oil stains underwater, which is detrimental to its antifouling property. On the other hand, the PDMS/PZS and PDMS/HSS-PZS surfaces can significantly inhibit the adhesion of oil stains and fouling organisms, exhibiting preferable superoleophobicity on different

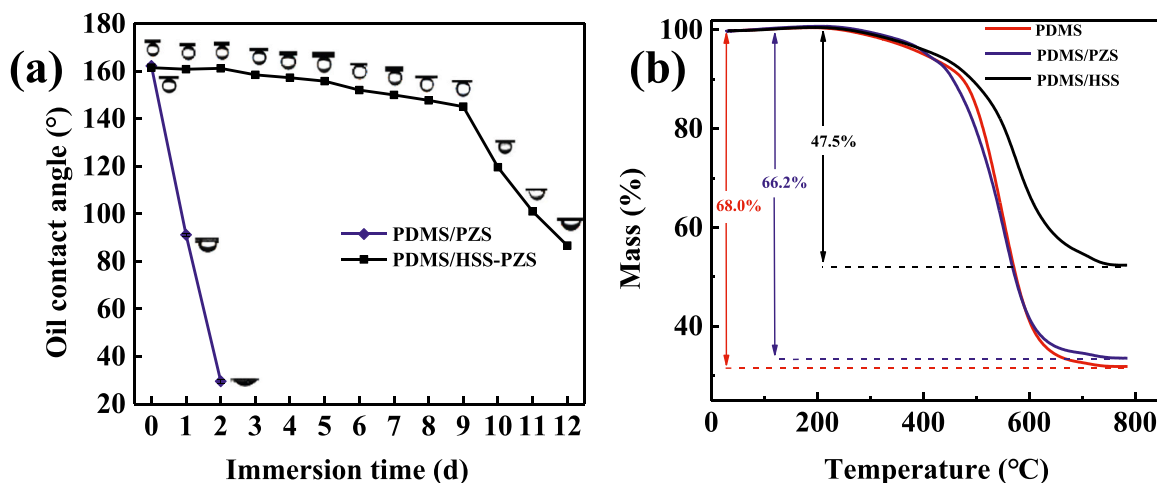


Fig. 8. (a) OCAs of different coating specimens over immersion time; (b) TGA of PDMS, PDMS/PZS and PDMS/HSS coatings.

substrates.

3.4. Stability of PDMS/HSS-PZS coating

Although the PDMS/PZS and PDMS/HSS-PZS coatings can effectively prevent the adhesion of oil stains on different substrates, their durability is still in question. The stability of coatings is prerequisite for their long-term antifouling and corrosion resistance. To evaluate the durability and stability, the surfaces of PDMS/PZS, PDMS/HSS and PDMS/HSS-PZS samples were checked after immersion tests in 3.5% NaCl solution for different time, and the bonding strength of PDMS/HSS and PDMS/HSS-PZS on AA6063 was investigated in 30 d test (Fig. 7a-b). In addition, we also measured the bonding strength of PDMS/HSS and PDMS/HSS-PZS to Al matrix at different temperatures (Fig. 7c). As shown in Fig. 7(a), many cracks appear on the surface of PDMS/PZS coating after immersion in 3.5% NaCl solution for 1 d, and the coating even peels off partially after 2 d and massively after 3 d, suggesting that the PDMS/PZS coating is poor in bonding strength. By contrast, no cracks appear on the surface of PDMS/HSS and PDMS/HSS-PZS surfaces even after 30 d of immersion, which is ascribed to the excellent blocking barrier of superhydrophobic PDMS/HSS surface against seawater. Meanwhile, the bonding strength of PDMS/HSS and PDMS/HSS-PZS only decrease slightly after 30 d of immersion, remaining above 1 MPa and 0.9 MPa, respectively (Fig. 7b), demonstrating excellent durability and stability. Furthermore, the bonding strength of PDMS/HSS and PDMS/HSS-PZS coatings shows no decreasing tendency in the temperature range from 0 °C to 100 °C (Fig. 7c), further proving the superb stability of PDMS/HSS and PDMS/HSS-PZS surfaces in 3.5% NaCl solution.

To evaluate the mechanical stability of the composite coating, OCAs versus abrasion and peeling cycles are shown in Fig. 7d-e. It can be found that the OCAs of PDMS/HSS-PZS decrease with increasing abrasion cycles (Fig. 7d), but they still sustain above 151° after 100 cycles of abrasion. Their mechanical stability is further proven by the tape-peeling test, as shown in Fig. 7e. After 200 peeling cycles, the OCAs still sustain above 141.3°, indicating that the PDMS/HSS-PZS coating has excellent scratch resistance.

From Fig. 6, it is seen that the PDMS/HSS coating shows strong lipophilicity, but the PDMS/PZS and PDMS/HSS-PZS coating exhibit excellent oil repellency, however, their long-term oil repellency properties are still unknown. Therefore, we investigated the OCA of PDMS/PZS and PDMS/HSS-PZS after long-term immersion in simulated seawater, as shown in Fig. 8a. Before soaking, the underwater OCAs of PDMS/PZS and PDMS/HSS-PZS coatings are 162.3° and 161.5°, respectively, demonstrating excellent superoleophobicity. However, the OCA of PDMS/PZS drops sharply to 91.2° after only 1 d of soaking. After

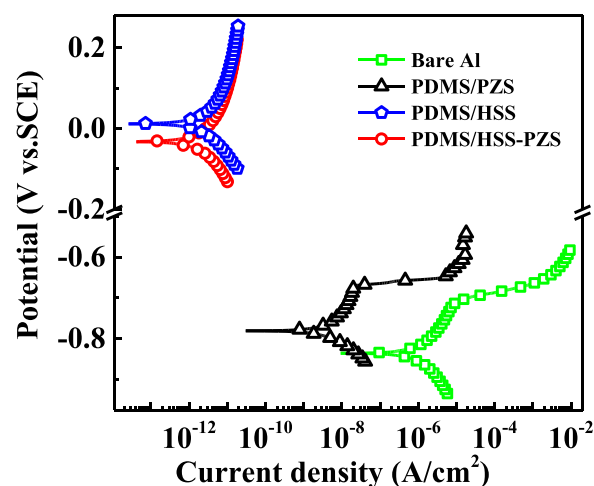


Fig. 9. Polarization curves of different coating specimens in 3.5 wt% NaCl solution.

2 d, the PDMS/PZS even becomes oleophilic possibly due to surface structural degradation caused by the hydroscopic and peeling coating. In contrast, the PDMS/HSS-PZS surface is superoleophobic with the OCA above 150° even after 7 d of soaking, indicating extraordinary durability in oil repellency. Namely, the PDMS/HSS-PZS coating exhibits better long-term oil repellency than the PDMS/PZS. Fig. 8b shows the TGA curves of the above three coatings, compared with the PDMS, the PDMS/HSS and PDMS/PZS present less weight loss, demonstrating that the addition of HD-SiO₂ and PAA-ZnO NPs effectively improves the thermal stability of PDMS based coatings.

3.5. Corrosion resistance measurements

The potentiodynamic polarization tests were used to estimate the corrosion rate of AA6063 aluminum substrate underneath the above coatings. Fig. 9 shows the Tafel curves of the bare, PDMS/PZS, PDMS/HSS and PDMS/HSS-PZS coated Al electrodes. Before testing, all specimens have been immersed in 3.5 wt% NaCl solution for 30 min Table 2 lists the relevant electrochemical parameters including E_{corr} and i_{corr} calculated by Tafel fitting method. Not surprisingly, the bare Al alloy has the highest i_{corr} (15.7 $\mu\text{A}/\text{cm}^2$) and the most negative E_{corr} (−0.836 V vs. SCE), indicating that the bare Al alloy suffers from severe corrosion. For the PDMS/PZS specimen, its E_{corr} slightly shifts positively along with its i_{corr} decreasing slowly to 6 nA/cm^2 . Furthermore, the E_{corr} of PDMS/HSS

Table 2

Corrosion potential (E_{corr}) and corrosion current density (i_{corr}) of different specimens.

Specimens	E_{corr} (V vs.SCE)	i_{corr} (A/cm ²)
Bare Al	-0.836	1.57×10^{-5}
PDMS/PZS	-0.781	6.43×10^{-9}
PDMS/HSS	0.011	4.46×10^{-12}
PDMS/HSS-PZS	-0.033	4.63×10^{-12}

and PDMS/HSS-PZS specimens significantly shifts positively. Particularly, the i_{corr} of PDMS/HSS and PDMS/HSS-PZS specimens decrease by seven orders of magnitude compared with that of bare Al alloy electrode, demonstrating excellent corrosion resistance on seawater. Table S1 also compares the corrosion current density (i_{corr}) of the PDMS/HSS-PZS coated Al alloy in this work with other superwettable coatings from literatures. According to their inhibitive efficiency, it can be found that the PDMS/HSS-PZS coating exhibits excellent anti-corrosion performance on Al substrate among all other coatings.

EIS has been widely recommended as a non-destructive technique to

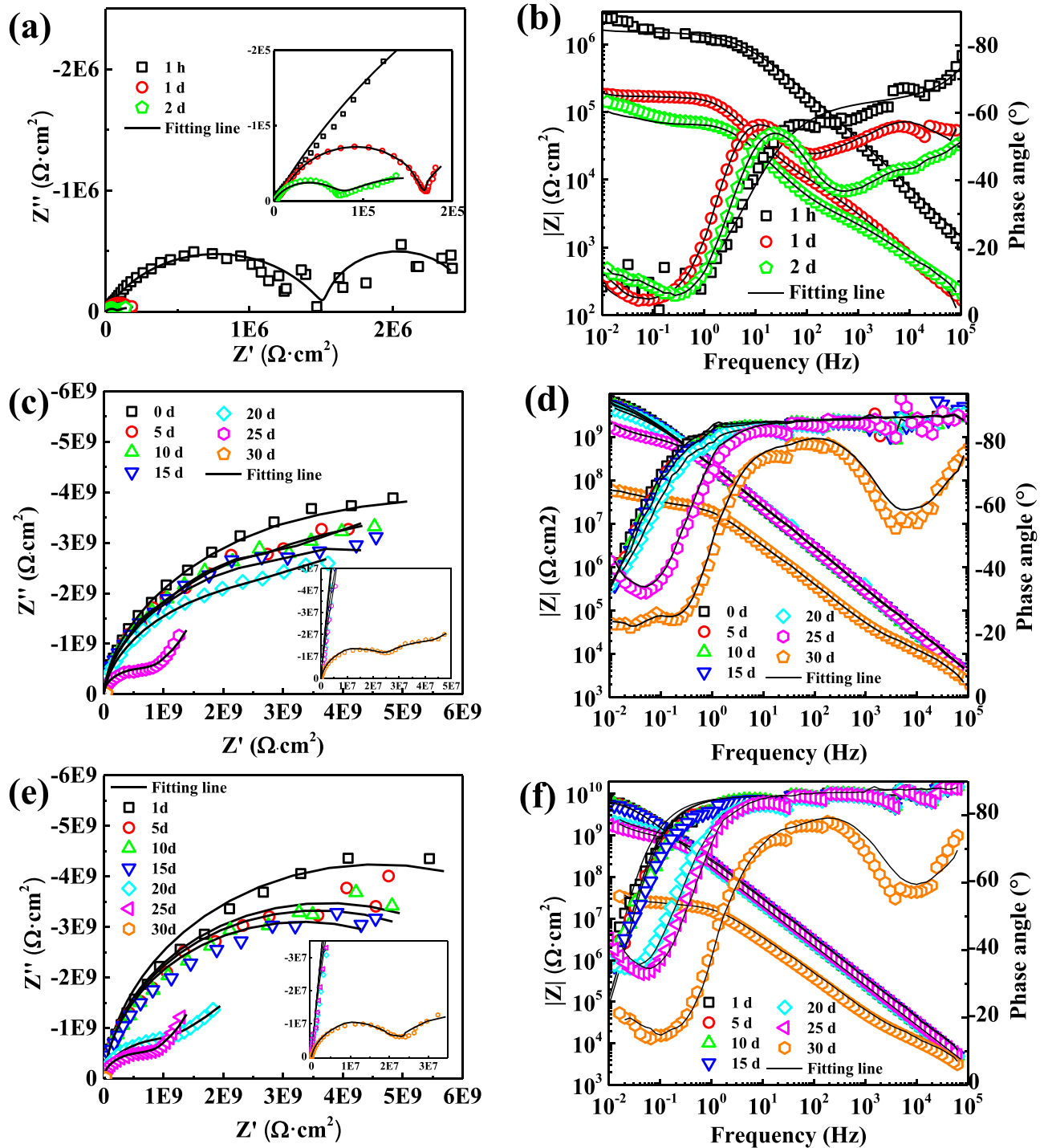


Fig. 10. Nyquist and Bode plots after different immersion time in 3.5 wt% NaCl solution for: (a) and (b) PDMS/PZS; (c) and (d) PDMS/HSS; (e) and (f) PDMS/HSS-PZS coated Al alloy electrodes.

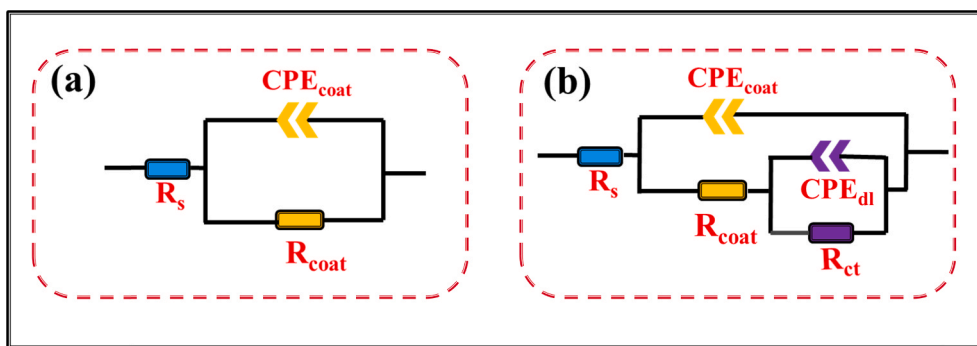


Fig. 11. Electrical equivalent circuits for EIS of coated specimens during immersion periods: (a) PDMS/HSS in 1–20 d and PDMS/HSS-PZS in 1–15 d, (b) PDMS/HSS in 25–30 d, PDMS/HSS-PZS in 20–30 d and PDMS/PZS in the whole immersion period.

evaluate protective properties of coating, which is usually conducted through a small amplitude sinusoidal excitation [51]. Fig. 10 summarizes the Nyquist and Bode plots of PDMS/PZS, PDMS/HSS and PDMS/HSS-PZS coated specimens after immersion in 3.5 wt% NaCl solution for different periods. From the Nyquist plots, it is seen that these coating specimens are different regarding their time-dependence impedance. The PDMS/PZS has two capacitive loops, corresponding to the coating semicircle at high frequency and the double electric layer at medium or low-frequency regions, respectively. This demonstrates that the electrochemical corrosion of Al matrix occurs even at the beginning of immersion, which is attributed to strong hydroscopicity of superoleophobic PDMS/PZS coating. Both PDMS/HSS-PZS and PDMS/HSS have only one capacitive loop during 15 d to 20 d of immersion, respectively, suggesting that no detectable corrosion occurs during this period thanks to the durable blocking barrier of superhydrophobic layer.

From the Bode plot in Fig. 10, it can be clearly seen that the impedance modulus $|Z|$ of all coating samples decreases with increasing immersion time. However, the impedance modulus at 0.01 Hz (denoted as $|Z|_{0.01 \text{ Hz}}$) of PDMS/HSS and PDMS/HSS-PZS are much higher than that of PDMS/PZS, mainly attributed to the extraordinary corrosion resistance of superhydrophobic PDMS/HSS layer. In particular, the $|Z|_{0.01 \text{ Hz}}$ values of PDMS/HSS and PDMS/HSS-PZS are still above $10^9 \Omega \text{ cm}^2$ even after 25 d of immersion, thanks to the durable and robust barrier of superhydrophobic PDMS/HSS layer. In contrast, the $|Z|_{0.01 \text{ Hz}}$ value of PDMS/PZS dropped sharply just after 1 d of immersion.

According to their EIS characteristics at different periods, two electrical equivalent circuits (ECs) in Fig. 11 are presented to fit the EIS in Fig. 10, in which a single time constant EC in Fig. 11a is employed to fit the EIS with single semi-circles, while a two time-constant EC (Fig. 11b) to fit the EIS with two semi-circles, where R_s represents the solution resistance, C_{coat} and R_{coat} the coating capacitance and coating resistance, C_{dl} and R_{ct} the double-layer capacitance and charge transfer resistance between the coating and the Al matrix, respectively. For all ECs, constant phase elements (CPE) are used to replace capacitances due to the inhomogeneous current distribution on the electrode surfaces [52]. In the early stage of immersion (for PDMS/HSS: from 1 d to 20 d; for PDMS/HSS-PZS: from 1 d to 15 d), there is only a single semicircle associated with the coating resistance R_{coat} and capacitance C_{coat} instead of the R_{ct} and C_{dl} , which is possibly ascribed to that the excellent barrier of the coatings inhibits early ingress of corrosive species (Cl^- ions, H_2O , etc.). That is, no apparent corrosion occurs at the coating/Al alloy interface, a simple EC in Fig. 11a is therefore much suitable. Nevertheless, in the later stage (PDMS/HSS: from 25 d to 30 d; PDMS/HSS-PZS: from 20 d to 30 d), corrosive species would have finally permeated into the coating/Al interface, leading to the electrochemical dissolution of Al matrix corresponding to the second semicircle in their EIS, and a two time-constant EC in Fig. 11b is more suitable.

The impedance modulus at 0.01 Hz is often used as an important index to assess the overall corrosion resistance of coatings. The $|Z|_{0.01 \text{ Hz}}$

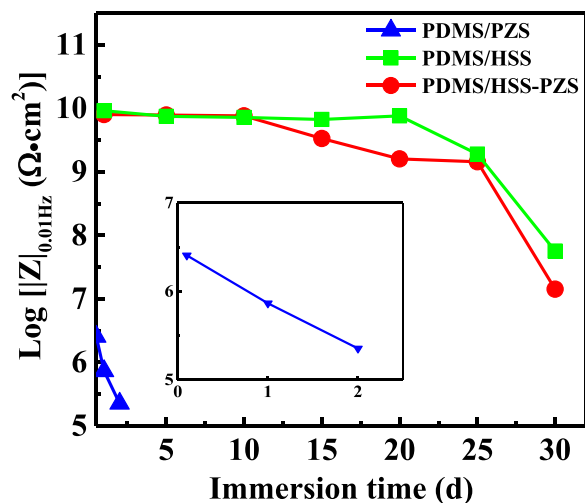


Fig. 12. The change of impedance modulus at 0.01 Hz ($|Z|_{0.01 \text{ Hz}}$) of different coatings over immersion time.

value of each coated specimen over immersion time is plotted in Fig. 12. It can be found that the PDMS/PZS coating provides good protection with the $|Z|_{0.01 \text{ Hz}}$ of $2.547 \text{ M}\Omega \text{ cm}^2$ in the early soaking. However, the $|Z|_{0.01 \text{ Hz}}$ value drops to $151.7 \text{ k}\Omega \text{ cm}^2$ sharply after only 2 d, suggesting a complete failure in their barrier effect. In contrast, the PDMS/HSS and PDMS/HSS-PZS specimens show as high $|Z|_{0.01 \text{ Hz}}$ values as $9552 \text{ M}\Omega \text{ cm}^2$ and $8948 \text{ M}\Omega \text{ cm}^2$ in the initial immersion stage, respectively. Moreover, their $|Z|_{0.01 \text{ Hz}}$ values maintain at above $10^9 \Omega \text{ cm}^2$ even in 25 d immersion, and only reduce to $43.62 \text{ M}\Omega \text{ cm}^2$ after 30 d, demonstrating excellent anticorrosion durability of the PDMS/HSS coating, which might be attributed to that superhydrophobic PDMS/HSS-PZS functions as an inner layer to inhibit the corrosion of Al substrate by water repellency and itself physical shielding effect.

In addition, the tribocorrosion behavior of PDMS/HSS-PZS was conducted by using a tribocorrosion tester under 0.5 MPa pressure. Fig. 13 shows the EIS of PDMS/HSS-PZS coating after different friction cycles. From the Nyquist plot, the PDMS/HSS-PZS coated specimens have only one capacitive loop before friction, and the semicircle diameter shrinks gradually with prolonged friction cycles. After 140 cycles, the Nyquist plot presents two capacitive loops with a new one appearing at low frequency region. The corresponding $|Z|_{0.01 \text{ Hz}}$ value also gradually decreases with friction cycles, but it still maintains at above $10^9 \Omega \text{ cm}^2$ after 120 friction cycles. Even after 180 friction cycles, the $|Z|_{0.01 \text{ Hz}}$ still keeps above $10^7 \Omega \text{ cm}^2$, indicating that the PDMS/HSS-PZS coating can resist moderate tribocorrosion.

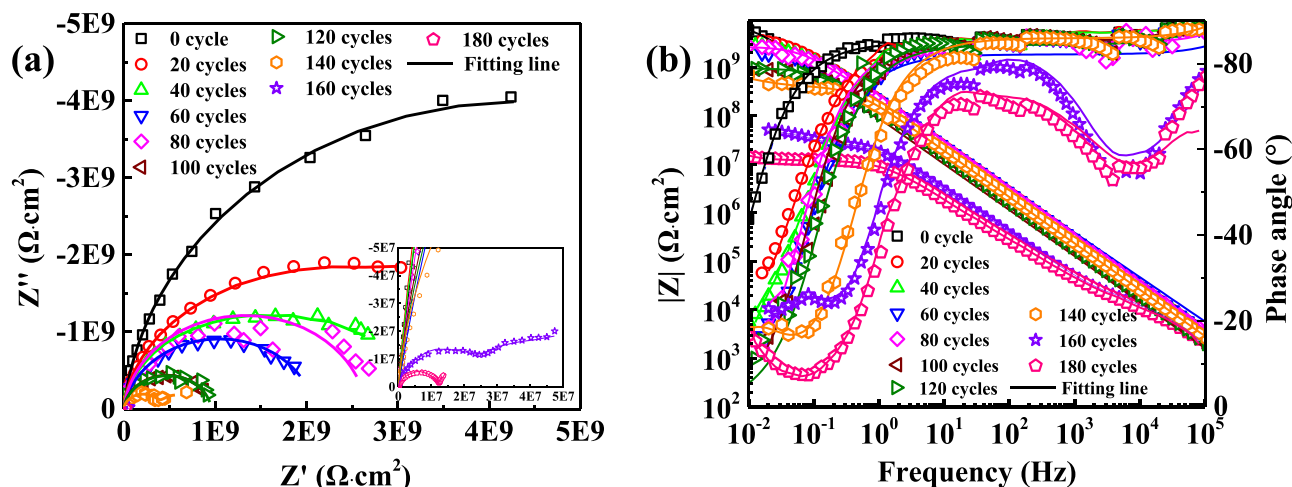


Fig. 13. Nyquist and Bode plots after different immersion time in 3.5 wt% NaCl solution for PDMS/HSS-PZS after different friction cycles.

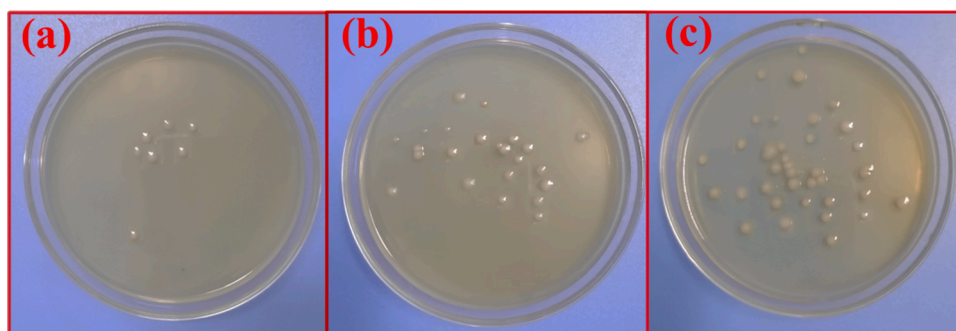


Fig. 14. Growth status of *Pseudomonas* sp. colonies on different coating specimens after 3 d culture: (a) PDMS/HSS-PZS; (b) PDMS/HSS; (c) bare Al.

Table 3

Colony number on different coatings and antibacterial efficiency after 3 d culture.

Group	The numbers of colony	Antibacterial efficiency
Bare Al	37	–
PDMS/HSS	23	37.8%
PDMS/HSS-PZS	7	81.1%

3.6. Characterization of antibacterial and antifouling properties

Although the PDMS/PZS coating exhibits excellent oil repellency, its durability and corrosion resistance were much inferior to the PDMS/HSS and PDMS/HSS-PZS coatings, therefore, only the latter two was selected for antibacterial and antifouling tests. Before tests, all the specimens were sterilized in high-temperature autoclave in advance, and then soaked in *Pseudomonas* sp. (10^5 cell/mL) culture medium at 28 °C for 3 d. Ultimately, after the *Pseudomonas* sp. suspension was diluted at a certain ratio, the colonies were counted to obtain the number of bacteria attached to the surface of bare Al, PDMS/HSS and PDMS/HSS-PZS coating for antibacterial efficiency estimate. The bacteria colonies are photographed in Fig. 14 with the antibacterial rate in Table 3. It can be found that not only the superhydrophobic PDMS/HSS surface but also the superoleophobic PDMS/HSS-PZS surface have promising antibacterial effects. In detail, the PDMS/HSS coating has an antibacterial efficiency of 37.8% compared with the bare Al. This is possibly ascribed to that the superhydrophobic PDMS/HSS surface reduces the contact area of culture medium with the coating and refrains the biofilm adhesion of *Pseudomonas* sp. Nonetheless, the superhydrophobic PDMS/HSS surface is oleophilic underwater and the lipid of bacteria tended to adhere to the

oleophilic surface, which reduces severely the antibacterial ability of the superhydrophobic PDMS/HSS coating. However, the superoleophobic PDMS/HSS-PZS coating presents a higher antibacterial efficiency (81.1%) than the PDMS/HSS, which is attributed to that the coating can prevent the adhesion of biofilm by capturing a layer of water molecules; in addition, the implanted ZnO particles can also refrain bacterial reproduction. Therefore, it is the synergistic effect of ZnO and superoleophobic surface that endows the PDMS/HSS-PZS coating excellent antibacterial ability.

To estimate the antifouling performance of the coatings, *Chlorella vulgaris* was inoculated in culture dishes with bare Al and PDMS/HSS specimens placed in the bottom of the dishes, respectively. As shown in Fig. 15a, with increasing culture time, the concentration of *Chlorella vulgaris* increases, and *Chlorella vulgaris* in the mediums containing bare Al and PDMS/HSS specimens grow much quicker than that in the medium containing PDMS/HSS-PZS specimen. The PDMS/HSS-PZS coating refrains the growth of *Chlorella vulgaris*, not only in the medium but also on the coating surface, which is possibly due to that the gradual release of ZnO from the PDMS/HSS-PZS coating suppresses the reproduction of *Chlorella vulgaris*. From Fig. 15b, it can be found that although the growth status of *Chlorella vulgaris* in the culture medium with bare Al sample and PDMS/HSS sample are almost identical after 14 d of cultivation, the number of *Chlorella vulgaris* attached to the PDMS/HSS surface is slightly less than that to the bare Al surface, which was due to that the superhydrophobic PDMS/HSS reduces the contact area between the coating and the aqueous medium. In addition, it should be noted that the coverage ratio of *Chlorella vulgaris* attached to the PDMS/HSS-PZS coating reaches 10.1%, which is less than that of the bare Al and PDMS/HSS samples (81.4%), as seen in Table S2, indicating that the coating containing ZnO particles has the best antibiofouling

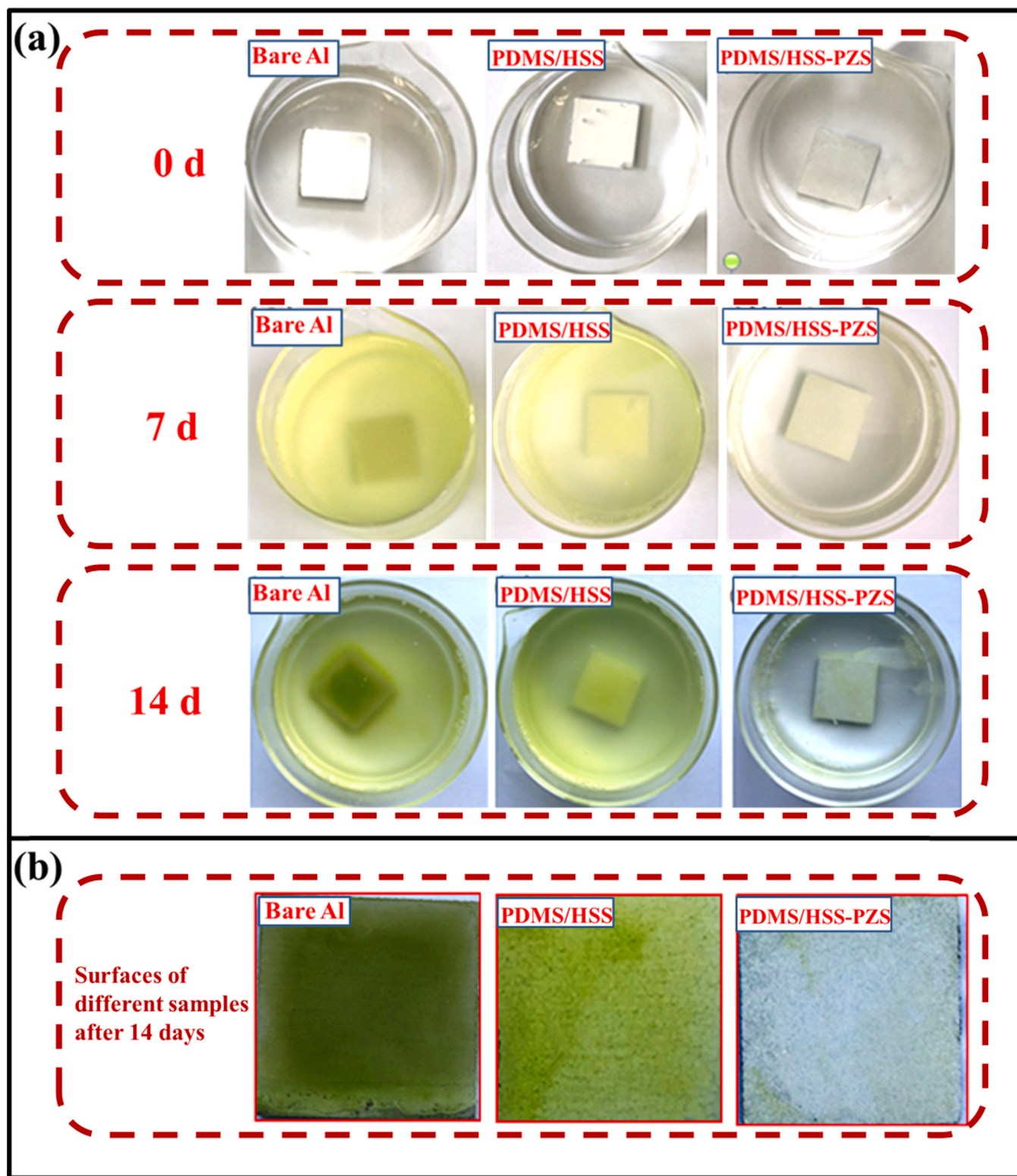


Fig. 15. Photos of (a) *Chlorella vulgaris* growth in culture medium with different coating specimens soaked in; (b) *Chlorella vulgaris* attachment to different coating surfaces.

performance.

The PDMS/HSS-PZS composite coating presents a relatively low affinity to oil stains underwater thanks to the excellent oil repellency of its outer layer, which can prevent the contact and adhesion of the oil stains and biofilm by capturing a water layer. In addition, the outer bactericidal ZnO particles also play a key role as bactericides. Moreover, the

superhydrophobic inner layer of PDMS/HSS-PZS could inhibit the corrosion of Al substrate by water repellency. Therefore, it is the synergistic effect that endows the PDMS/HSS-PZS coating with long-term antifouling and anti-corrosion ability in 3.5% NaCl solution. It is worth noting that PDMS/HSS-PZS coating is also eco-friendly thanks to the employment of low surface energy hexadecyltrimethoxysilane

(HDTMS), avoiding the environmental and biological impacts by use of fluorinated compounds such as perfluoropolyether [38,53,54]. Therefore, it is supposed that the PDMS/HSS-PZS composite coating may be one of the competitive candidates for marine coatings due to its low-cost fabrication technique.

4. Conclusions

We reported a feasible technique to synthesize superoleophobic and superhydrophobic composite coating to achieve excellent antifouling, waterproof and anticorrosion properties. The surface morphology and wettability of the coating can be adjusted by the PDMS to PAA-ZnO mass ratio, and it can reach the maximum OCA of 160.4° when the mass ratio is 2:1. The composite coating could effectively prevent the attachment of oil stain underwater by capturing water layer on the surface. In addition, it exhibits excellent antibacterial ability thanks to the synergistic effect of superoleophobicity and bactericide (ZnO).

Electrochemical tests demonstrate that the PDMS/HSS-PZS composite coating inhibits the corrosion rate of Al alloy by 7 orders of magnitude even after 25 d of immersion in 3.5 wt% NaCl, exhibiting excellent anti-corrosion performance. Antifouling test indicates that the coating could reduce the adhesion of oily molecules underwater and then refrain the formation of *Pseudomonas* sp. and *Chlorella vulgaris* biofilm. The promising durability on anti-fouling and anti-corrosion of the PDMS/HSS-PZS composite coating, is possibly ascribed to that the PDMS/HSS superhydrophobic inner layer inhibits the ingress of seawater, and the PDMS/PZS outer layer effectively prevents the attachment of fouling organisms and bacteria. In addition, the composite coating could be fabricated on large scale by spraying method due to its simplicity, low cost, and universal adaptability to various substrates.

CRediT authorship contribution statement

Xian Chao Xia & Xiang Kang Cao: Conceptualization, Methodology, Writing – Original draft preparation, Data curation, both authors contributed equally to this work. **Guang Yi Cai & Dan Jiang & Fei Zhang:** Investigation, Data curation. **Ze Hua Dong:** Writing – Reviewing, Supervision and Editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.colsurfa.2021.127323](https://doi.org/10.1016/j.colsurfa.2021.127323).

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